

Table 1: Differences between vector and raster maps

	Raster maps	Vector maps
Created by	Created with pixels	Created with mathematical formulae
Scaling	Quality loss when scaling is done	Scalable to any size
Programs used	Adobe Photoshop, Corel Painter, Gimp, Art-weaver, Pixlr X	Adobe Illustrator, Sketch, Coreldraw, Affinity Designer, Inkscape
Comprise	Raster data is made up of pixels (also grid cells). These are usually regularly spaced and square	Real-world features in maps are represented as points, lines, and polygons (areas)
File types	.jpg, .png, .gif, .tiff, .psd, .bmp	.ai, .eps, .cdr, .svg, .pdf, .dxf
Advantages	The inherent nature of raster maps, e.g., one attribute maps, is ideally suited for mathematical modelling and quantitative analysis.	Topology rules can help data integrity with vector data models. Network analysis and proximity operations use such an approach.
Drawback	Raster maps reflect only one attribute or characteristic for an area. Processing of diverse characteristics becomes cumbersome	Vector data is processing intensive. Any feature edits require updates on topology. With many features, vector manipulation algorithms are complex

resources analysis support systems (GRASS GIS), system for automated geoscientific analyses (SAGA GIS) and MapWindow. Machine learning owes its popularity to open source programming/GUI choices like WEKA, KNIME, R and Python programming. The merger of GIS and machine learning has opened new avenues in research to explore beyond what is visible through sensors.

In this article we are using the Python console of the quantum GIS platform — PyQGIS.

Maps are available either in raster or vector form for study and we can choose them depending on our task, software and region of interest. Vector data is focused on modelling discrete features with precise shapes and boundaries. Raster data is more about modelling continuous phenomena of the earth and images. The same can be imported to the *Project* template of the QGIS window.

The map used in the present setup is downloaded from <https://www.naturalearthdata.com>. Natural Earth Vector comes in ESRI shapefile format, the *de facto* standard for vector geodata. Character encoding is UTF-8. Natural Earth Raster comes in TIFF format with a TFW world file. All

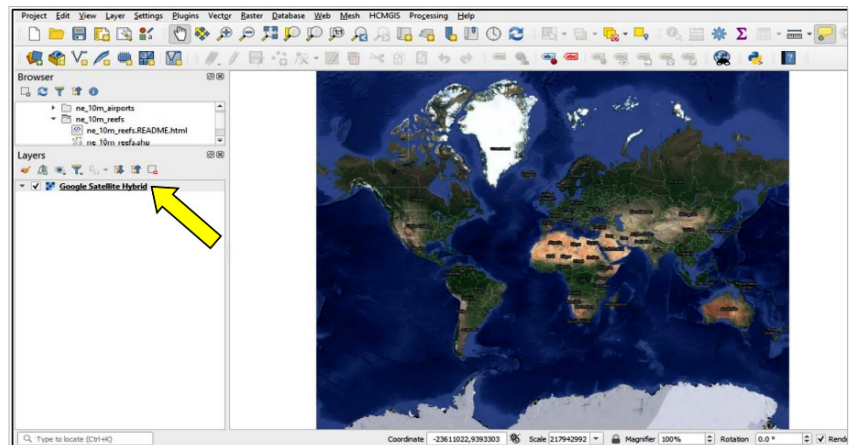


Figure 2: Project view of QGIS

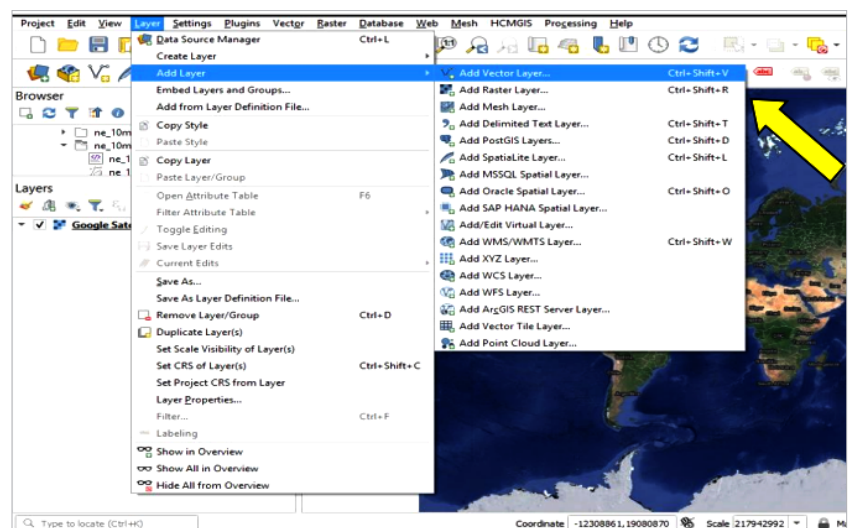


Figure 3: Layer selection in QGIS